

Topic 16: Kinetics II

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include investigating different methods for tracking rates of reaction by gas collection or change in mass or colour change, investigating clock reactions such as Harcourt-Essen, using a simple rates experiment to calculate an activation energy.

Mathematical skills that could be developed in this topic include plotting and justifying the shapes of rate-concentration and concentration-time graphs, calculating half-life of a reaction, calculating activation energy from a suitable graph, rearranging the Arrhenius equation in the form $y = mx + c$.

Within this topic, students can consider different methods used to measure reaction rates and collect valid data. Through the analysis of this data, and a knowledge of rate equations, they can see how chemists are able to propose models to describe the mechanisms of chemical reactions.

Students should:

1. understand the terms:
 - i rate of reaction
 - ii rate equation
 - iii order with respect to a substance in a rate equation
 - iv overall order of reaction
 - v rate constant
 - vi half-life
 - vii rate-determining step
 - viii activation energy
 - ix heterogeneous and homogeneous catalyst
2. be able to determine and use rate equations of the form:
rate = $k[A]^m[B]^n$, where m and n are 0, 1 or 2
3. be able to select and justify a suitable experimental technique to obtain rate data for a given reaction, including:
 - i titration
 - ii colorimetry
 - iii mass change
 - iv volume of gas evolved
 - v other suitable technique(s) for a given reaction
4. understand experiments that can be used to investigate reaction rates by:
 - i an initial-rate method, carrying out separate experiments where different initial concentrations of one reagent are used
A 'clock reaction' is an acceptable approximation of this method
 - ii a continuous monitoring method to generate data to enable concentration-time or volume-time graphs to be plotted

Students should:

5. be able to calculate the rate of reaction and the half-life of a first-order reaction using data from a concentration-time or a volume-time graph
6. be able to deduce the order (0, 1 or 2) with respect to a substance in a rate equation using data from:
 - i a concentration-time graph
 - ii a rate-concentration graph
7. be able to deduce the order (0, 1 or 2) with respect to a substance in a rate equation using data from an initial-rate method
8. understand how to:
 - i obtain data to calculate the order with respect to the reactants (and the hydrogen ion) in the acid-catalysed iodination of propanone
 - ii use these data to make predictions about species involved in the rate-determining step
 - iii deduce a possible mechanism for the reaction
9. be able to deduce a rate-determining step from a rate equation and vice versa
10. be able to deduce a reaction mechanism, using knowledge from a rate equation and the stoichiometric equation for a reaction
11. understand that knowledge of the rate equations for the hydrolysis of halogenoalkanes can be used to provide evidence for S_N1 or S_N2 mechanisms for tertiary and primary halogenoalkane hydrolysis
12. be able to use graphical methods to find the activation energy for a reaction from experimental data
The Arrhenius equation will be given if needed.

CORE PRACTICAL 13a and 13b: Rates of reaction

Following the rate of the iodine-propanone reaction by a titrimetric method and investigating a 'clock reaction' (Harcourt-Esson, iodine clock)

CORE PRACTICAL 14: Finding the activation energy of a reaction